

TUNKU ABDUL RAHMAN UNIVERSITY OF MANAGEMENT AND TECHNOLOGY

FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY

ACADEMIC YEAR 2024/2025

OCTOBER EXAMINATION

BACS2163 SOFTWARE ENGINEERING

THURSDAY, 10 OCTOBER 2024

TIME: 9.00 AM – 11.00 AM (2 HOURS)

BACHELOR OF COMPUTER SCIENCE (HONOURS) IN DATA SCIENCE
BACHELOR OF COMPUTER SCIENCE (HONOURS) IN INTERACTIVE SOFTWARE
TECHNOLOGY

BACHELOR OF INFORMATION TECHNOLOGY (HONOURS) IN SOFTWARE SYSTEMS
DEVELOPMENT

BACHELOR OF SOFTWARE ENGINEERING (HONOURS)

Instructions to Candidates:

Answer **ALL** questions. All questions carry equal marks.

BACS2163 SOFTWARE ENGINEERING**Question 1**

- a) Y2J is interested in creating library management system based on the **FOUR (4)** key layers of layered software engineering technology. Describe each layer based on the system. (12 marks)

- b) In Software project management, a project manager would need to consider factors that could motivate employees throughout the project. Describe any **THREE (3)** levels of needs in Maslow's motivation model. (9 marks)

- c) Differentiate between software engineering and system engineering. (4 marks)

[Total: 25 marks]

Question 2

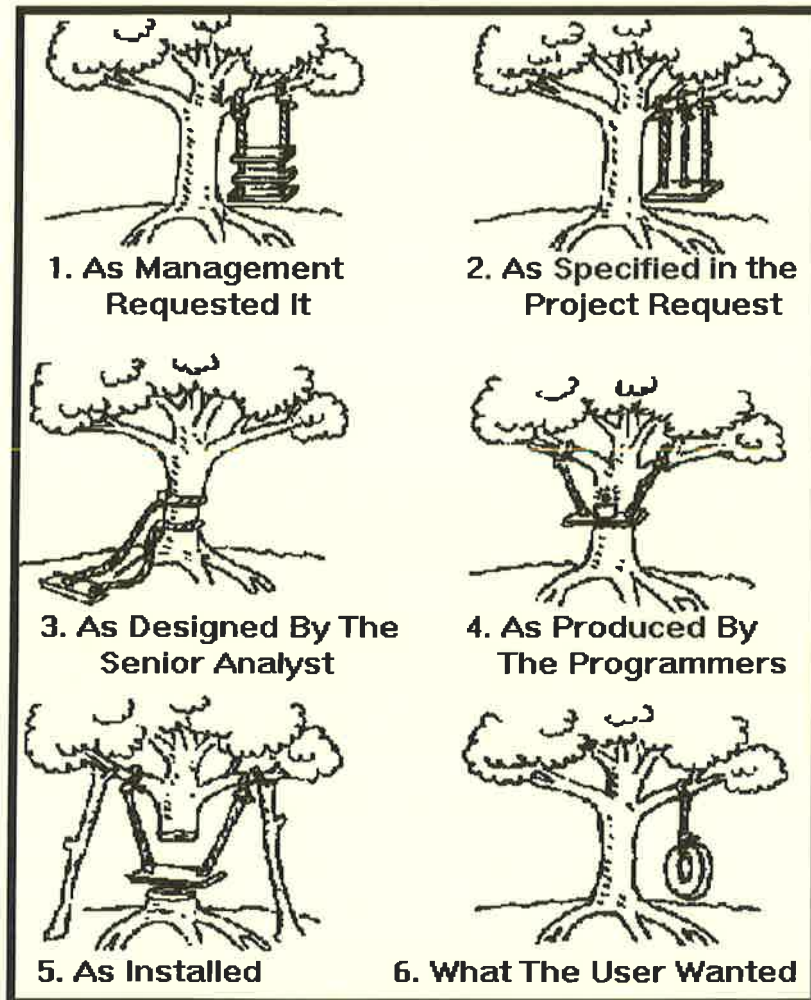
- a) Company KT is planning to develop an online document management system called ITS portal that is used by internal staff to store, share, and work on some confidential documents. You, and IT project manager, are responsible to manage this project with senior software engineers.
 - (i) Identify any **TWO (2)** advantages and **TWO (2)** disadvantages of using viewpoint-oriented analysis method in requirements gathering and analysis. (4 marks)
 - (ii) Identify any **SIX (6)** viewpoints of ITS portal and prepare a viewpoint hierarchy. (12 marks)

- b) The Neato Robotics vacuum cleaner, with its advanced algorithm and sensors, adapts to clean narrow spaces, avoid obstacles like toys, and automatically recharge when low on battery, ensuring thorough cleaning without interruptions. Compose and explain 3 sets of stimuli and associating response for the Neato Robotics system. (9 marks)

[Total: 25 marks]

BACS2163 SOFTWARE ENGINEERING**Question 3**

a) Answer the following questions based on given Figure 1.



Source: <http://www.robelle.com/library/smugbook/tree.gif>

Figure 1: Software Development Process

- (i) Analyse the scenario above and with **THREE (3)** points explain why a software process model is necessary for software project development. You may make any relevant assumptions to support your response. (6 marks)
- (ii) Identify the **FOUR (4)** software process activities involved in completion of software development project. (12 marks)
- (iii) Do you agree that nowadays, many software development companies are utilising agile? Justify your answers with **THREE (3)** agile software development principles. (7 marks)

[Total: 25 marks]

BACS2163 SOFTWARE ENGINEERING**Question 4**

- a) Based on Figure 2, suggest and explain **SIX (6)** user interface design principles that must be included in developing the interface of the login page. (12 marks)

The screenshot shows the Amtrak.com website interface. At the top is a navigation bar with links: Home, Timetables, Stations v, Routes v, Deals v, Plan v, and Guest Rewards v. Below the navigation bar is a yellow error message box with a warning icon: "Sorry for the inconvenience. Currently there is a system issue preventing your request from being processed. [Error ID: 5045]". Below the error message is the "PASSENGER INFORMATION" section, which includes a "Log In" heading. The text below the heading says: "If you already have an Amtrak.com user profile, log in here. If you don't have a user profile, continue as a guest by completing the information in the next section." Below this text are two input fields: "User ID" and "Password". To the right of the "Password" field is a "Login" button with a right-pointing arrow.

Source: <https://www.google.com/search?q=mobile+game+kids>

Figure 2: Login page

- b) You, as a project manager have been assigned to lead project team for setting up a Human Resource system. New York City acts as the main hub controlling global HR rules. Regional offices in London, Tokyo, and Sydney act as branches handling local HR tasks while following the central rules. This system ensures HR practices are consistent worldwide yet adaptable to regional needs. It eases communication, ensures policy consistency, and can grow as the company expands globally. Recommend and draw an appropriate system organisation model for the above project. Explain the model and justify your recommendation. (13 marks)

[Total: 25 marks]